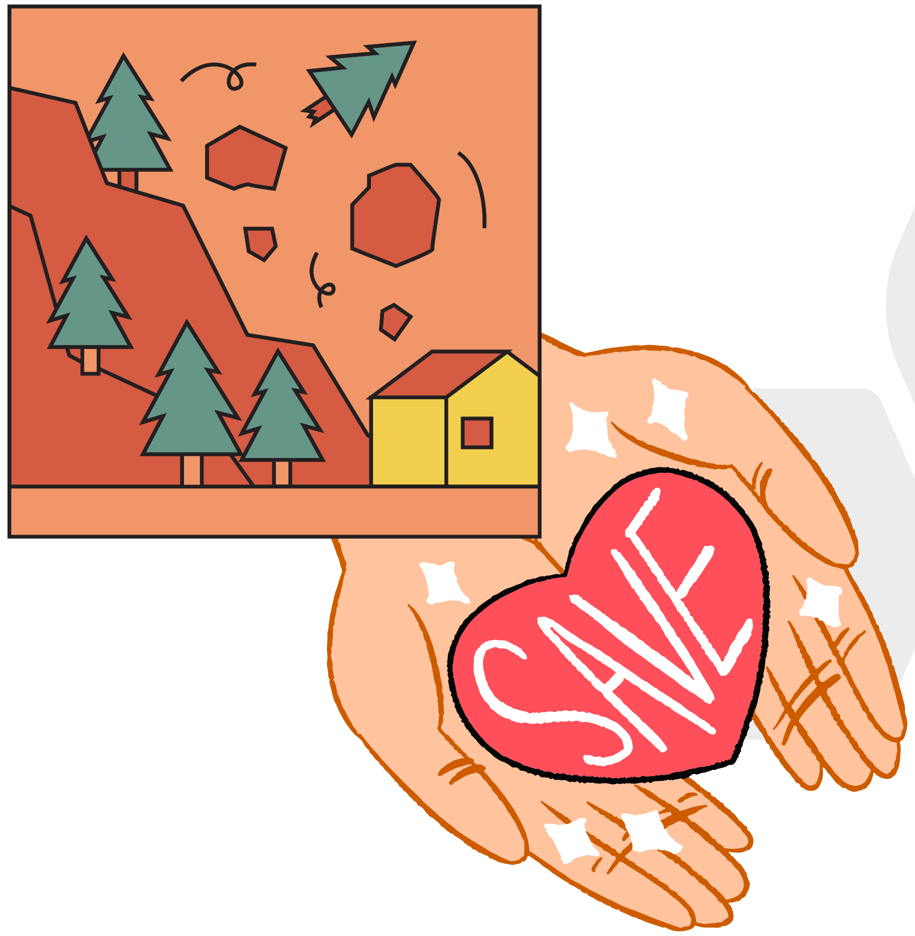
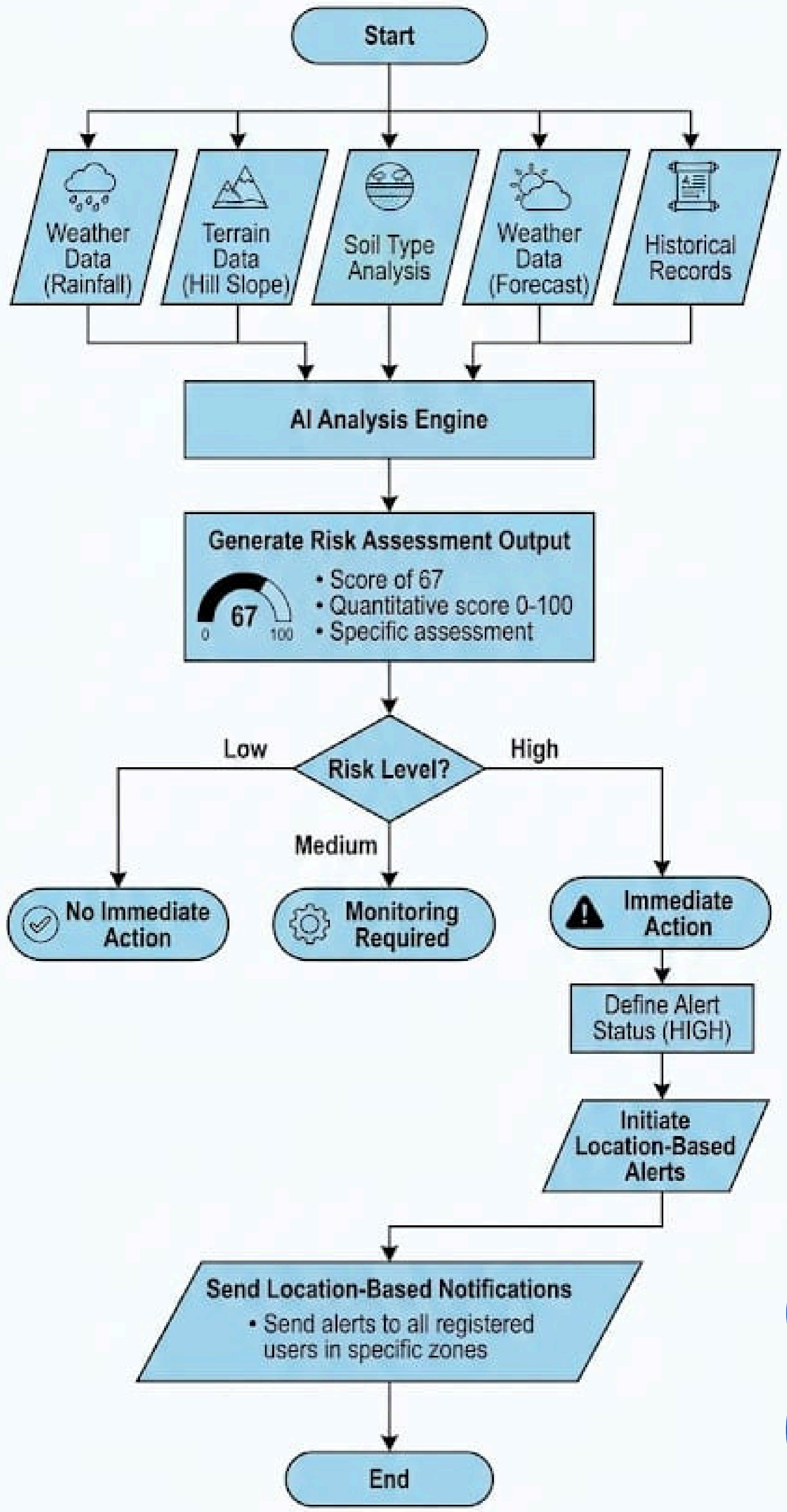
SMART INDIA HACKATHON 2026

TITLE PAGE

- **PROBLEM STATEMENT ID- SIH26001**
- **PROBLEM STATEMENT- AI-BASESD EARLY WARNING AND LANDSLIDE RISK MONITORING SYSTEM IN NER**
- **THEME- DISASTER MANAGEMENT**
- **PS CATEGORY- SOFTWARE**
- **TEAM ID-**
- **TEAM NAME- SCAPEGOATS**



The Process of Landslide Risk Prediction



IDEA TITLE

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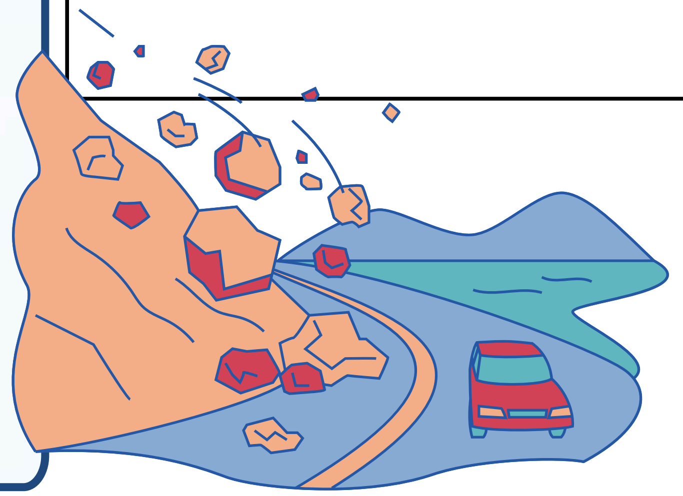
Landsafe AI — Intelligent Landslide Risk Prediction & Early Warning System

INOVATIONS

- Multi-factor risk analysis
- Rainfall intensity/duration
- Slope/elevation
- Soil/geological information
- Historical landslide data
- Weather conditions

SOLUTIONS

- Continuously analyses environmental and geographical factors.
- Generates a location-specific landslide risk score.
- Displays vulnerable areas on an interactive map.
- Sends early warnings when risk crosses a threshold.



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TECHNICAL APPROACH

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DATA SOURCES

↓

WEATHER / RAINFALL

TERRAIN & ELEVATION

SOIL / GEOLOGICAL DATA

HISTORICAL LANDSLIDES

↓

DATA PREPROCESSING

↓

FEATURE EXTRACTION

↓

AI / ML RISK MODEL

↓

RISK SCORE

↓

GIS-BASED RISK MAP

↓

ALERT GENERATION

↓

USER / AUTHORITY

TECHNOLOGIES USED

FRONTEND

- REACT / HTML-CSS-JS

BACKEND

- PYTHON
- FASTAPI/FLASK

AI/ML

- PYTHON
- SCIKIT-LEARN
- XGBOOST/RANDOM FOREST

MAPS

- GOOGLE MAPS / OPENSTREETMAP
- GIS LAYERS

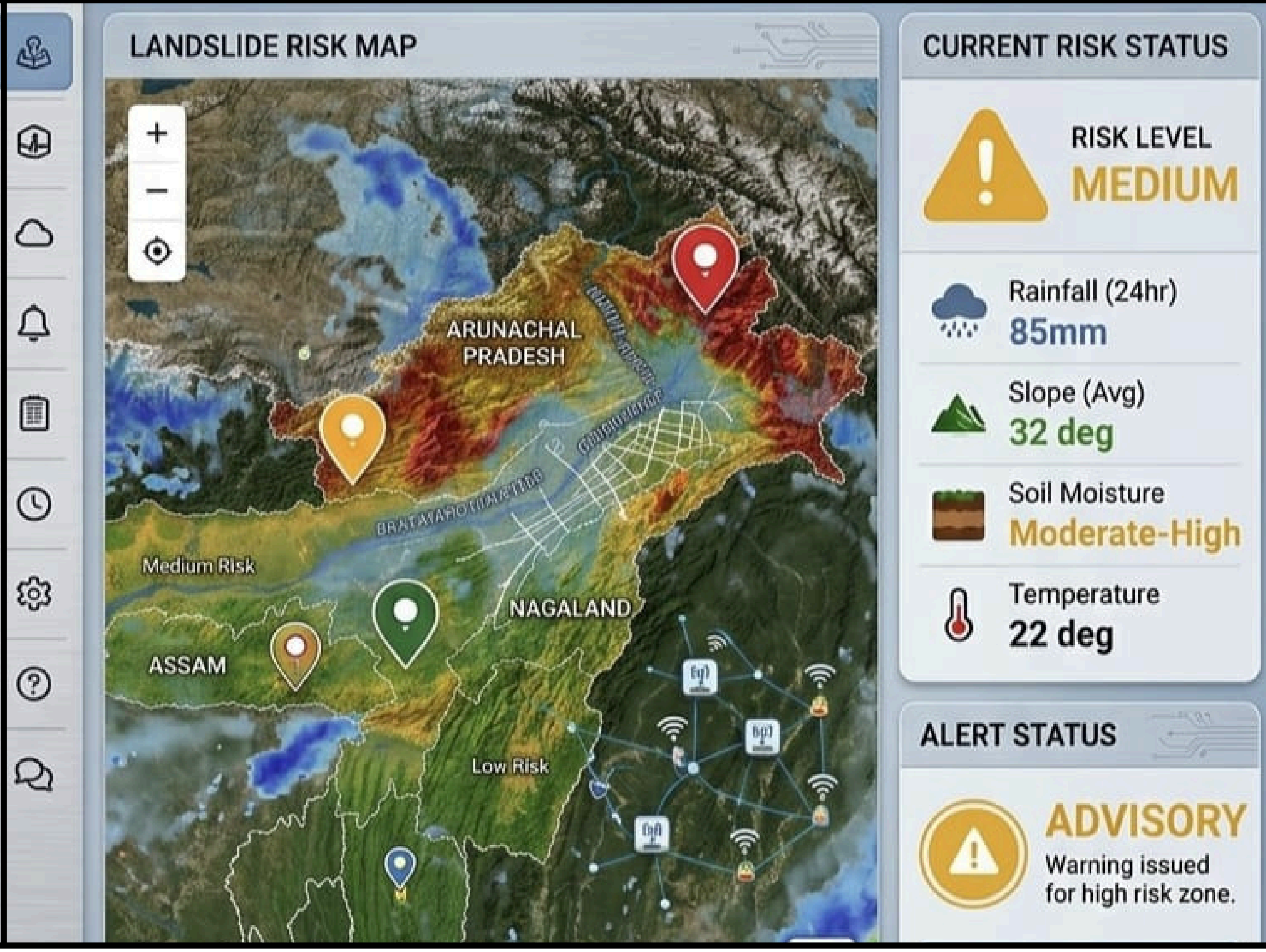
DATABASE

- FIREBASE / POSTGRESQL

NOTIFICATIONS

- FIREBASE CLOUD MESSAGING / SMS

Early Warning System



FEASIBILITY AND VIABILITY

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SOFTWARE-FIRST IMPLEMENTATION

- Uses publicly available environmental/geographical datasets.
- ML model can initially be trained using historical data.
- No expensive physical infrastructure required for the prototype.
- Can begin with a limited geographical region and scale later.

⚠️ CHALLENGES

Data availability

→ Different regions have different terrain/soil characteristics.

False alarms

→ Incorrect predictions could reduce user trust.

Limited real-time data

→ Some environmental datasets aren't updated continuously.

Model generalisation

→ A model trained for one region may not work equally well elsewhere.

🛠️ SOLUTIONS

Data availability

→Combine multiple public datasets and allow region-specific calibration.

False alarms

→Use multiple factors instead of rainfall alone and provide risk levels rather than absolute predictions.

Real-time limitations

→Design the system to work with the latest available data and clearly display data freshness.

Generalisation

→Retrain/calibrate models using regional historical data.

IMPACT AND BENEFITS

BENIFITS

🚨 Early Warning
More time for evacuation and emergency response.

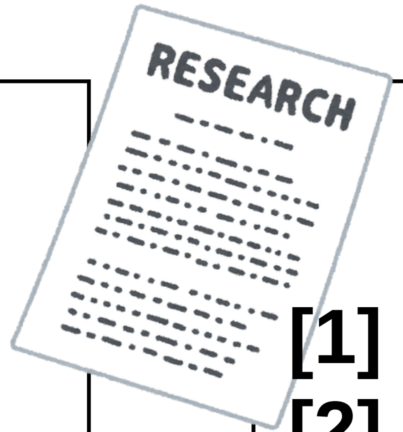
🗺️ Risk Visualization
Authorities can identify high-risk areas quickly.

💰 Reduced Economic Loss
Helps protect homes, infrastructure and roads.

🌱 Environmental Protection
Supports better monitoring of vulnerable terrain.

📱 Public Awareness
People can see the risk level for their location.

RESEARCH AND REFERENCES



Research / Data Sources

- Historical landslide datasets
- Rainfall/weather datasets
- Digital Elevation Models (DEM)
- Geological/soil datasets
- Government disaster-management reports
- Research papers on landslide susceptibility prediction
- Existing landslide early-warning systems

REFERENCES

[1] ISRO/NRSC — Landslide Atlas of India
[2] ISRO/NRSC — National Database for Emergency Management (NDEM)
[3] Sikkim Himalaya — Remote sensing and GIS-based landslide susceptibility mapping
[4] Sikkim — Rainfall thresholds for real-time landslide early warning
[5] Northeast India — Machine Learning-based Landslide Susceptibility Modeling in Dibang Valley